

Pandas Series Practice Problems

Based on Water Quality Monitoring Data

Instructions

1. Read the CSV file using `pd.read_csv()` and extract individual columns as Series.
2. Answer all questions using Pandas Series operations.
3. Show your code and output for each question.

Data Loading

```
import pandas as pd

# Load the data
df = pd.read_csv('water_quality.csv')

# Extract columns as Series
salinity = df['Salinity_(ppt)']
do = df['Dissolved_Oxygen_(mg/L)']
ph = df['pH_(standard_units)']
secchi = df['Secchi_Depth_(m)']
water_depth = df['Water_Depth_(m)']
water_temp = df['Water_Temp_(?C)']
air_temp = df['AirTemp_(C)']
year = df['Year']
site = df['Site_Id']
field_tech = df['Field_Tech']
```

Series Creation & Basic Operations

- Q1 Create a Series named `do_readings` containing the first 50 Dissolved Oxygen measurements.
- Q2 Create a Series `temp_difference` by subtracting `air_temp` from `water_temp`.
- Q3 Set a custom index for the `salinity` Series using the row numbers and name the Series `'Salinity_Levels'`.
- Q4 Create a Series from a dictionary where keys are unique site IDs and values are the average pH for each site.
- Q5 Access the 10th, 25th, and 50th elements of the `ph` Series using both `iloc` and `loc` (after setting a custom index).

Series Attributes

- Q6 Find the `dtype`, `shape`, `size`, and `ndim` of the `salinity` Series.
- Q7 Check if the `secchi` Series contains any NaN values using the `hasnans` attribute.
- Q8 Extract the values of the `water_depth` Series as a NumPy array.
- Q9 Create a Series with missing values from the `do` column and demonstrate the use of `isnull()` and `notnull()`.
- Q10 Print the `index` and `name` attributes of the `site` Series after setting an appropriate name.

Data Inspection Methods

- Q11 Use `head()` and `tail()` to display the first 10 and last 5 records of the `water_temp` Series.
- Q12 Use `sample()` to randomly select 20 records from the `ph` Series and calculate their mean.
- Q13 Use `sample()` with `frac=0.15` and `random_state=42` to create a representative sample of the `salinity` Series.
- Q14 Use `value_counts()` to find the most common `Site_Id` in the dataset.
- Q15 Display the frequency distribution of `Field_Tech` with percentages.

Sorting Operations

- Q16 Sort the `water_temp` Series in descending order and display the top 10 highest temperatures.
- Q17 Sort the `do` Series in ascending order and display the bottom 10 dissolved oxygen readings.
- Q18 Sort the `site` Series alphabetically.
- Q19 Sort the `year` Series in-place using `sort_values(inplace=True)`.
- Q20 Sort the `field_tech` Series and observe how it handles duplicate values.

Data Cleaning & Missing Values

- Q21 Create a Series `salinity_with_na` where salinity is set to NaN for readings before 1990.
- Q22 Use `dropna()` to remove all NaN values from the Series created in Q26.
- Q23 Use `fillna()` to replace NaN values in the `secchi` Series with the median Secchi depth.
- Q24 Identify the number of missing values in the `pH` column.
- Q25 Show how to handle missing values in the `air_temp` Series by filling with the mean air temperature.

Advanced Operations

- Q26 Create a Boolean Series `high_do` that is True where Dissolved Oxygen exceeds 8 mg/L.
- Q27 Use boolean indexing to filter the `water_temp` Series where `water_depth` $\geq$ 1 meter.
- Q28 Create a new Series `adjusted_salinity` where salinity is increased by 10% for readings with `pH` $\geq$ 7.
- Q29 Use `value_counts()` on the `year` Series to find the year with the most measurements.
- Q30 Create a Series `oxygen_saturation` where $\text{saturation} = \text{do} / (14.6 - 0.394 * \text{water_temp})$ and handle any division by zero.

Aggregation & Statistical Operations

- Q31 Calculate the `mean`, `median`, `mode`, `min`, and `max` of the `ph` Series.
- Q32 Compute the standard deviation and variance of the `water_temp` Series.
- Q33 Find the 25th, 50th, and 75th percentiles of the `salinity` Series using `quantile()`.
- Q34 Calculate the total number of measurements and the average Secchi depth using Series methods.
- Q35 Create a summary statistics table for the `do` Series using `describe()`.

String Operations (for categorical columns)

- Q36 Convert all values in the `site` Series to uppercase.
- Q37 Create a Boolean Series that is True when the `field_tech` contains the letter 'e'.
- Q38 Use `str.len()` to find the length of each field technician name in the `field_tech` Series.
- Q39 Extract the first 4 characters of each site ID from the `site` Series.
- Q40 Count how many field technician names in the `field_tech` Series contain the substring 'Poe' or 'Phillips'.

DateTime Operations

- Q41 Convert the `Read_Date` column to datetime format and extract the month into a new Series `read_month`.
- Q42 Use `value_counts()` on the `read_month` Series to find which month has the most measurements.
- Q43 Create a Boolean Series `is_summer` that is True for readings from June, July, and August.
- Q44 Calculate the day of the week for each reading using `dt.day_of_week`.
- Q45 Filter the `water_temp` Series to include only readings from the year 2010 or later.

Combined Operations & Real-world Scenarios

- Q46 Create a Series `water_quality_index` where $WQI = do * 0.4 + pH * 0.3 + salinity * 0.3$.
- Q47 Calculate the average pH per site by creating a Series where index is site and value is mean pH.

- Q48 Find the top 5 sites with the highest average Dissolved Oxygen using Series operations.
- Q49 Create a Series `temp_rank` that ranks the `water_temp` values from highest to lowest.
- Q50 Determine the percentage of measurements with Dissolved Oxygen below 5 mg/L (hypoxic conditions).
- Q51 Create a Series that shows the cumulative sum of pH readings over the dataset (in original order).
- Q52 Use `value_counts()` and `sort_values()` together to find the most frequently occurring pH value.
- Q53 Create a Series `adjusted_temp` where water temperature is converted to Fahrenheit using $F = C * 9/5 + 32$.
- Q54 Calculate the average Secchi depth per site using grouping (create separate Series for each site).
- Q55 Create a pivot-like result: a Series showing the average temperature per year.

Challenge Questions

- Q56 Create a Series `water_quality_classification` where each reading is classified as:
- 'Excellent': DO ≥ 8 AND pH BETWEEN 7 AND 8.5
 - 'Good': DO ≥ 6 AND pH BETWEEN 6.5 AND 9
 - 'Poor': otherwise
- Q57 Calculate the running (cumulative) average of Dissolved Oxygen and find the point where it drops below 7 mg/L.
- Q58 Create a Series showing the year-over-year change in average water temperature.
- Q59 Find the measurements that have both high salinity (≥ 3 ppt) and low DO (≤ 5 mg/L) using boolean operators.
- Q60 Create a Series that shows the top 3 years by average pH for each site.

Solution Template

For each question, provide your solution in this format:

```
# Q1: Solution
# Code here
# Output here
```